

Ming-Lung (Stanley) Tsai

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EDUCATION

National Taiwan University

Sep 2023 - Present

B.S. in Computer Science and Information Engineering

Taipei, Taiwan

- Member of NTU's 18th Creativity and Entrepreneurship Program cohort; developed and market-tested CouPro through its product-building and validation curriculum.

HONORS AND SELECTED PROGRAMS

- Third Place, NTU Creativity and Entrepreneurship Program (CouPro), 2026
- Selected for Silicon Valley participation, NTU AI Builders Challenge, 2026 (offer received)
- First Place, Cyber Security Skills Golden Shield Awards
- Top Five, Cyber Security Skills Golden Shield Awards
- 2026 TeamT5 Security Camp
- Advanced Information Security Summer School (AIS3)
- MOE Information Security Incubation Program (ISIP) - Cybersecurity Camp

RESEARCH EXPERIENCE

Undergraduate Researcher, SSLab | kvTZ-Trusty

2025 - Present

- Extending Prof. Shih-Wei Li's kvTZ architecture to Android Trusty through pKVM and AVF, targeting virtualized TrustZone services on physical devices.

LLM Multi-Agent Bystander Study

2026

- Designed controlled LLM multi-agent experiments on whether diffused responsibility causes a bystander effect; compared intervention across persona-driven scenarios and experimental conditions.

SELECTED TECHNICAL PROJECTS

US Stock-Market NLP

2024 - 2025

- Applied hierarchical clustering, RNNs, and BERT to annual-report MD&A text to predict year-over-year financial performance; co-developed a confidence-weighted ensemble that outperformed its standalone components.

Nested pKVM and SMMU Device Passthrough

2026

- Extended a collaborative Linux/pKVM kernel tree with guest Stage-2 device-passthrough support; built separate validation paths for protected-pVM DMA enforcement and full-Ubuntu e1000e I/O.

CouPro | CTO and Product Lead

2026

- Architected CouPro's DevOps workflow and release process across the Next.js web frontend, Django backend, and mobile frontend.
- Introduced agentic AI into development and authored team playbooks for technical discussion, implementation, and documentation.
- Maintained all three application layers; shipped three market iterations in two months, incorporating field feedback into each release.

LEADERSHIP AND SERVICE

NTU Homeless Service Club | Vice President, 5th and 6th Terms

- Established NGO partnerships, produced campus exhibitions, and co-led the club's first print journal, extending its outreach to local markets.

TECHNICAL SKILLS AND METHODS

Languages: Python, C/C++, JavaScript, TypeScript, SQL, Bash, PHP

Web and mobile: React, Next.js, React Native, Expo, Vue, Tailwind CSS

Backend and storage: Django/DRF, FastAPI, Laravel, PostgreSQL, SQLite, Redis, Cloudflare R2

Systems and security: Linux kernel, Android Trusty, KVM/pKVM, AVF, SMMU, QEMU, GDB, Burp Suite

AI and data: OpenAI/Gemini APIs, Ollama, PyTorch, NumPy, Pandas, Polars, SciPy, Matplotlib; NLP, BERT, RNNs, clustering, LLM multi-agent experiments

Testing and monitoring: pytest, Jest, Playwright, Maestro, Locust, Sentry, PostHog, Firebase Analytics

Development and automation: Git, GitHub Actions, Docker, Claude Code, Codex, Playwright CLI